

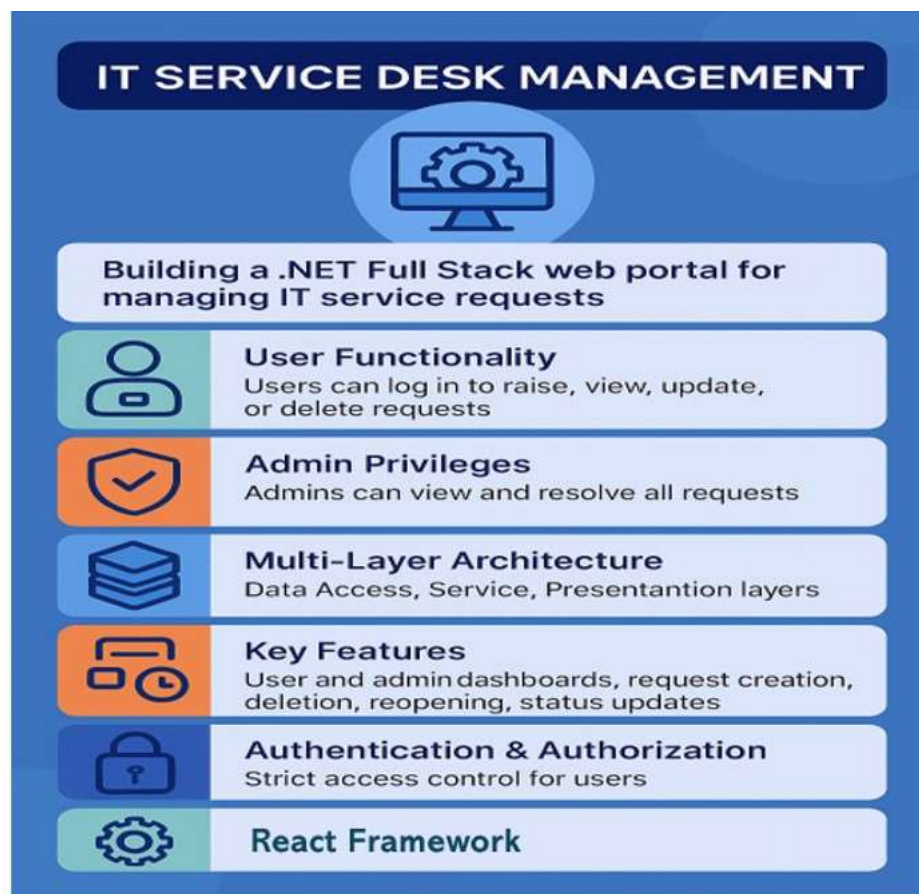
Capstone Project Usage Instructions. Read it carefully.

- The project/assessment will be available for the full project/assessment window.
- You may **start, pause, and continue** your assessment anytime during this window by using the **Save & Close** button.
- When you click **Save & Close**, your progress will be saved and you can return later to continue the assessment.
- After clicking **Save & Close**, if you log in again immediately, the **Resume** button may appear **disabled for about 7–8 minutes**.
 - This is normal; please wait a few minutes for the Resume button to become active.
- Do **not** click **Finish Test** until you are completely done with your project.
- Once you click **Finish Test**, your assessment will be **submitted permanently**, and you will **not** be able to log in or make any further changes.

Instructions

- Read all sections carefully: Ensure a thorough understanding of the requirements.
- Marks Allocation: Marks will be awarded for working functionalities only.
- Coding Standards: Adhere to coding standards as per Method1 to ensure a well-structured and maintainable solution.

Title: IT Service Desk Management



Problem Statement

Introduction:

An IT user logs into the IT Service Request portal to raise a service request for an issue they are experiencing. They can later revisit the

Objective:

The objective is to develop a user-friendly web portal that streamlines the management of IT service requests within the organization. Through this application, employees can raise and track issues, while administrators can efficiently view, manage, and resolve them.

Design Specifications:

The Application will follow the design specification approach outlined below:

1. Use multi-layer architecture (Data Access Layer, Service Layer, Presentation Layer)
2. Use Data Annotations to perform necessary validations.
3. Use Entity Framework Core to implement Data Access Layer
4. Use Web API to implement service Layer
5. Implement Authentication and Authorization mechanisms.
6. Use Azure Devops and Agile Methodology throughout the process
7. Apply structured exception handling.
8. Adhere to .NET Core coding standards and naming conventions.
9. Use SQLite for Data management .

Database : Configuration for Sqlite

1. Open Visual Studio -> Go to Extensions => Manage extensions-> browse for the option SQLite.



2. To complete the installation, and apply all the modification, visual studio should be closed.



1. Add the following packages using NuGet Package Manager:

- Microsoft.EntityFrameworkCore.Sqlite (version 9.0.9)
- Microsoft.EntityFrameworkCore.Tools (version 9.0.9)

2. Add a **Model** Folder and include the classes listed below

- Add a model class named **Role.cs** with the following properties.
 - RoleId (int, primary key, Identity(1,1))
 - RoleName (string, length(35), Required)
- Add another model class named **User.cs** with the following properties.
 - UserName (string,length(20), primary key)
 - Password (string, min-length(8), max-len(20), Required)
 - CreatedOn (DateTime.Now,Required)
 - RoleId (int, Navigation property from Role)
- Add a model class named **Status.cs** with the following properties.
 - StatusId (int, primary key, Identity(1,1))
 - Description (string, length(50), Required)
- Add another model class named **ServiceRequest.cs** with the following properties.
 - RequestId (int, primary key, Identity(1,1))
 - Description (string, length(50), Required)
 - Details (string, length(100), Required)
 - RaisedBy(string, length(20), Required)
 - RaisedOn (DateTime.Now,Required)
 - Justification(string, length(50), Required)
 - ReqStatus (int, Navigation property from status)

3. Add a folder "**Database**", To store database file (.db) .

4. Create a folder named **Repository** in the project and include an interface named **IRepository.cs** in it with the following specifications:

- public bool Authenticate(User user);
- public List<ServiceRequest> ViewRequests();
- public List<ServiceRequest> ViewRequests(string userName);
- public int RaiseRequest(ServiceRequest newRequest);
- public ServiceRequest GetRequestById(int requestId);
- public bool ReOpenRequest(ServiceRequest request);
- public List<ServiceRequest> GetRequestBySP(string userName);
- public bool CloseRequest(int requestId);
- public bool DeleteRequest(int requestId);
- public User GetUser(string userName);

5. Add a Repository class named **Repository.cs** which implements the **IRepository** Interface in Repository folder.

6. Add **HelpDeskDbContext.cs** class in Repository with below code

```

public HelpDeskDbContext(DbContextOptions<HelpDeskDbContext> options)
: base(options)
{
}

public class HelpDeskDbContextFactory :
IDesignTimeDbContextFactory<HelpDeskDbContext>
{
    public HelpDeskDbContext CreateDbContext(string[] args)
    {
        var options = new
DbContextOptionsBuilder<HelpDeskDbContext>()
        .UseSqlite("Data Source=Database\\ServiceDeskDB.db")
        .Options;
        return new HelpDeskDbContext(options);
    }
}

public DbSet<Role> Roles { get; set; }
public DbSet<User> Users { get; set; }
public DbSet<ServiceRequest> ServiceRequests { get; set; }
public DbSet<Status> Statuses { get; set; }
}

```

7. **Migration** : Run the necessary migration commands to add the following tables to the SQLite :

- Roles
- Users
- Status
- ServiceRequests

8. To View the tables and add master data, Go to Tools -> SQLite/Sql Server Compact Tool box, and add connection with sqlite database file(.db file).



Service Layer: Web API

1. Create a Web API controller named **ITSRPAPIController.cs** with the following endpoints and functionality:

- Authenticate User :


```

[HttpGet("Authenticate")]
public IActionResult Authenticate(string userName, string password)

```

 - Validates user login credentials
 - Returns NotFound if authentication fails
 - Returns Ok(user) on successful login
- Get All Service Requests


```

[HttpGet("GetAllRequest")]
public IActionResult GetAllRequest()

```

 - Retrieves a list of all service requests
 - Returns NotFound if list is empty
 - Returns Ok(List<ServiceRequest>)
- Get Requests by username

- Reopen an Existing Request

```
[HttpPost("reopen")]
public IActionResult ReOpenRequest(ServiceRequest request)
- Validates required fields
- Reopens the request in the system
- Returns ok with the updated service request
```

- Create a New Service Request

```
[HttpPost("CreateNewSerRequest")]
public IActionResult Post(ServiceRequest newRequest)
- Validates the new request
- Creates a new service request
- Returns CreatedAtAction with ID
```

- Get Service Request by ID

```
[HttpGet("GetRequestById")]
public ServiceRequest GetRequestById(int reqId)
- Fetches details of a specific request
- Returns the request object
```

- Get User Details

```
[HttpGet("GetUser")]
public IActionResult GetUser(string userName)
- Retrieves user information
- Returns NotFound if user does not exist
- Returns Ok(User)
```

- Close a Service Request

```
[HttpGet("CloseRequest")]
public IActionResult CloseRequest(int id)
- Marks the request as closed
- Returns NotFound if not found
- Returns Ok on success
```

- Delete a Service Request

```
[HttpGet("Delete")]
public IActionResult Delete(int id)
- Deletes an existing request
- Returns NotFound if deletion fails
- Returns Ok upon successful deletion
```

2. Update below code in Program.cs

- a. Add Before the code line , var app = builder.Build();

```

        Capstone.DAL ,
        "Database",
        "ServiceDeskDB.db"
    );
    builder.Services.AddDbContext<HelpDeskDbContext>(options =>
        options.UseSqlite($"Data Source={dbPath}"));

    builder.Services.AddDbContext<HelpDeskDbContext>();
    builder.Services.AddScoped<IRepository, Repository>();

    // ENABLE CORS
    var allowedOrigins = "_allowedOrigins";
    builder.Services.AddCors(options =>
    {
        options.AddPolicy(name: allowedOrigins,
            policy =>
            policy.AllowAnyOrigin().AllowAnyMethod().AllowAnyHeader());
    });

```

b. Add before the code line , app.Run();

```

//*****FOR CORS
app.UseCors(allowedOrigins);
//*****FOR CORS
app.MapControllers();

```

Note:

- Make sure, in launchSettings.json under the profile -> applicationUrl key, it should map to https url. (eg : "applicationUrl": <https://localhost:5193>)
- While running the API, if https security error come in browser, run the below command in command prompt.

/> dotnet dev-certs https --trust

Presentation Access Layer: React Framework

You must build the entire presentation layer using the React framework, including all UI components, screens, and interactions with proper state management and reusable components. React best practices should be followed such as component-based architecture, modular organization, and effective use of hooks to deliver a clean, responsive user experience.

Note: Install the necessary packages with below command in React Layer in command prompt

C:\CapstoneProject\clonefromDevOps\Capstone.React> **npm i axios react-router-dom**

Authentication : Only existing users who have been added to the Users table by the admin can access all the above screens. There is no permission for anonymous users to access any resource except the Login Screen

Expected Output:

Note: Learners are encouraged to use innovative and creative design approaches, provided they do not change the given captions or the required functionality.

User DemoGraphics

1. **Login Screen:** All fields are enabled. This is the default landing Screen for the application.

service requests raised by the currently signed in user.



The screenshot shows a web application interface with a header bar containing a 'Logout' link. Below the header, a welcome message 'Welcome rahul' is displayed, followed by a link to 'Raise New Request'. A table lists two service requests. The first request has ID 8, description 'H Card', requested by 'Rahul', creation date '5/21/2025 11:14:20 AM', and status 'New'. The second request has ID 9, description 'Card Install', requested by 'Rahul', creation date '5/21/2025 11:14:51 AM', and status 'Closed'. Both requests have links to 'Delete' and 'Re-Open'. At the bottom, a copyright notice reads 'Copyright © 2025 Accenture. All rights reserved.'

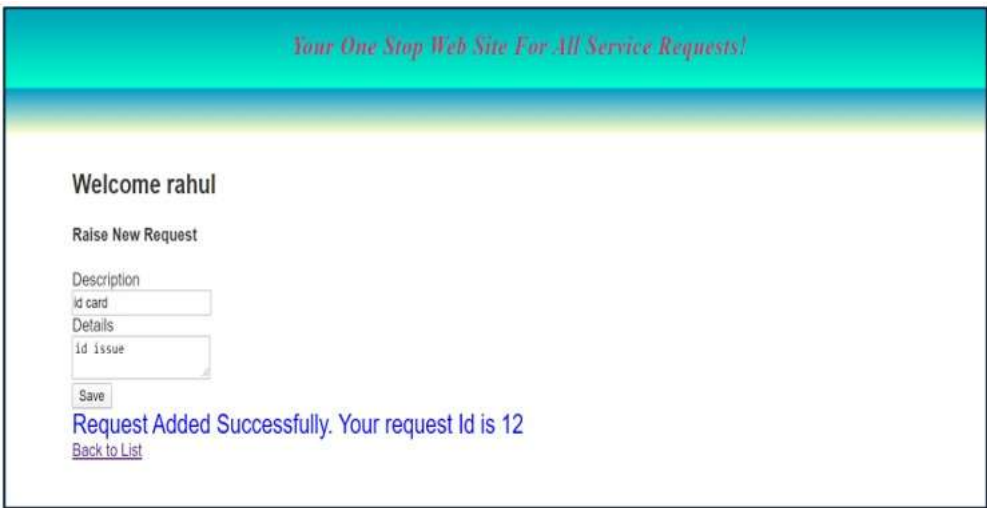
Request ID	Description	Requested By	Creation Date	Request Status
8	H Card	Rahul	5/21/2025 11:14:20 AM	New
9	Card Install	Rahul	5/21/2025 11:14:51 AM	Closed

3. **Add Service Request Demographics** : All fields are enabled. This screen is displayed when the user clicks on the **Raise New Request** Link in the home screen. Details about the service request will be added by the user. Upon successful authentication, the user must be redirected to the following screen:



The screenshot shows a web application interface with a header bar containing the text 'Your One Stop Web Site For All Service Requests!'. Below the header, a welcome message 'Welcome rahul' is displayed, followed by a link to 'Raise New Request'. The form contains two input fields: 'Description' and 'Details'. Below the 'Details' field is a 'Save' button and a 'Back to List' link. At the bottom, a copyright notice reads 'Copyright © 2025 Accenture. All rights reserved.'

The id of the request is displayed to the user when the details have been filled correctly and the **Save** button is clicked.



The screenshot shows the same web application interface as the previous one, but with the 'Description' field filled with 'Id card' and the 'Details' field filled with 'Id issue'. Below the 'Save' button, a message reads 'Request Added Successfully. Your request Id is 12'. The 'Back to List' link is still present. At the bottom, a copyright notice reads 'Copyright © 2025 Accenture. All rights reserved.'

4. All fields are read only. This screen allows a user to delete a specific request. This screen is displayed when the user clicks on the Delete link for a specific request record in the home screen.

Are you sure you want to delete this?

Description	Details	Creation Date
Id Card	Need An Id Card Holder	1/1/0001 12:00:00 AM

Delete Request

[Back to List](#)

A status message is displayed on the same screen on successful deletion.

5. Re-Open Request

This screen allows a user to re-open a closed request in case the user is not satisfied with the action taken on the request. All fields are read only, except the Justification Field. The Justification is mandatory to re-open a request. This screen is displayed when the user clicks on the Re-Open link for a specific request record in the home screen .

Your One Stop Web Site For All Service Requests!

Welcome

Re-Open Request

Description	Cant Install Sql	
Details	sql 2012 installation gives pipe error	
Creation Date	5/21/2020 11:14:51 AM	
Justification		

ReOpen Request

[Back to List](#)

Admin DemoGraphic:

1. Login Screen:

All fields are enabled. This is the default landing screen for the application.

USER LOGIN

EmailId

Password

LOGIN

Your One Stop Web Site For All Service Requests!

Welcome admin
[Search Requests](#)

Request ID	Description	Requested By	Creation Date	Request Status	
8	Id Card	Rahul	5/21/2020 11:14:29 AM	New	Close Request
9	Cant Install Sql	Rahul	5/21/2020 11:14:51 AM	Closed	
11	Access Rights	Kathryn	5/21/2020 1:02:33 PM	New	Close Request

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3. Search Request Demographics

All fields are enabled. This screen is displayed when the user clicks on the **Search Requests** Link in the home screen.

Your One Stop Web Site For All Service Requests!

Search Requests
 Enter the Requester Name:

The details of the request are displayed in a tabular format in the same screen as shown below:

Your One Stop Web Site For All Service Requests!

Search Requests
 Enter the Requester Name:

Request ID	Description	Details	Requested By	Creation Date	Request Status	
8	Id Card		Rahul	5/21/2020 11:14:29 AM	New	Close Request
9	Cant Install Sql		Rahul	5/21/2020 11:14:51 AM	Closed	

An appropriate message is shown if no match is found.

4. Close Request

When the admin clicks on the **Close Request** link for any specific request in the home screen, the status of that request is updated to closed and the Home screen is refreshed to show the updated change. There is no separate screen for this.

Home Screen before closing request with ID : 11

Welcome admin

[Search Requests](#)

Request ID	Description	Requested By	Creation Date	Request Status	
8	Id Card	Rahul	5/21/2020 11:14:29 AM	New	Close Request
9	Cant Install Sql	Rahul	5/21/2020 11:14:51 AM	Closed	
11	Access Rights	Kathryn	5/21/2020 1:02:33 PM	New	Close Request

Home Screen after closing request with ID : 11

Your One Stop Web Site For All Service Requests!

Welcome admin

[Search Requests](#)

Request ID	Description	Requested By	Creation Date	Request Status	
8	Id Card	Rahul	5/21/2020 11:14:29 AM	New	Close Request
9	Cant Install Sql	Rahul	5/21/2020 11:14:51 AM	Closed	
11	Access Rights	Kathryn	5/21/2020 1:02:33 PM	Closed	

Upload your response

Note:

File types permissible: .pdf,.doc,.docx,.zip

Maximum size of file can be upto 100MB.



Drag & drop file here

Or

+ Select File